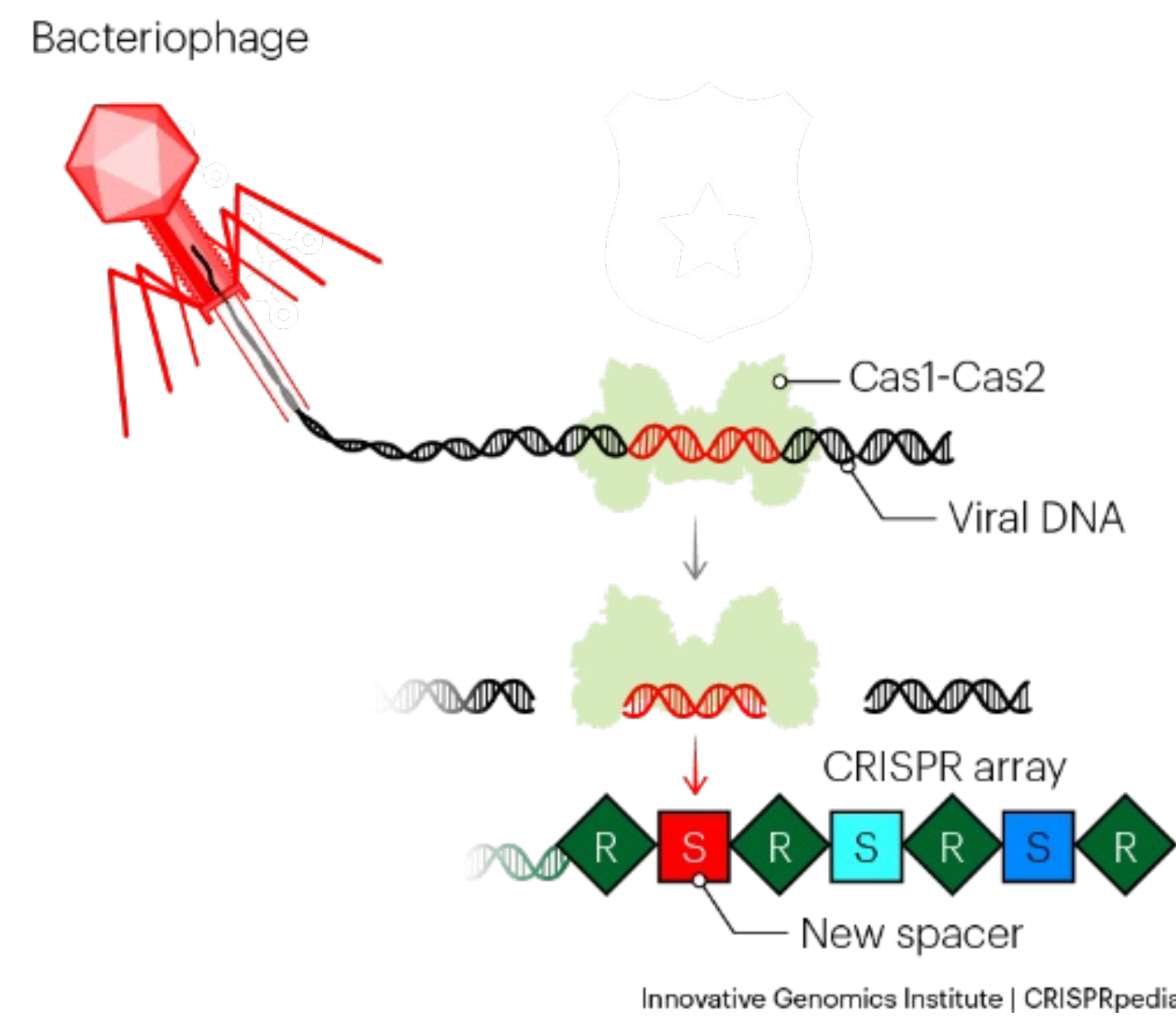


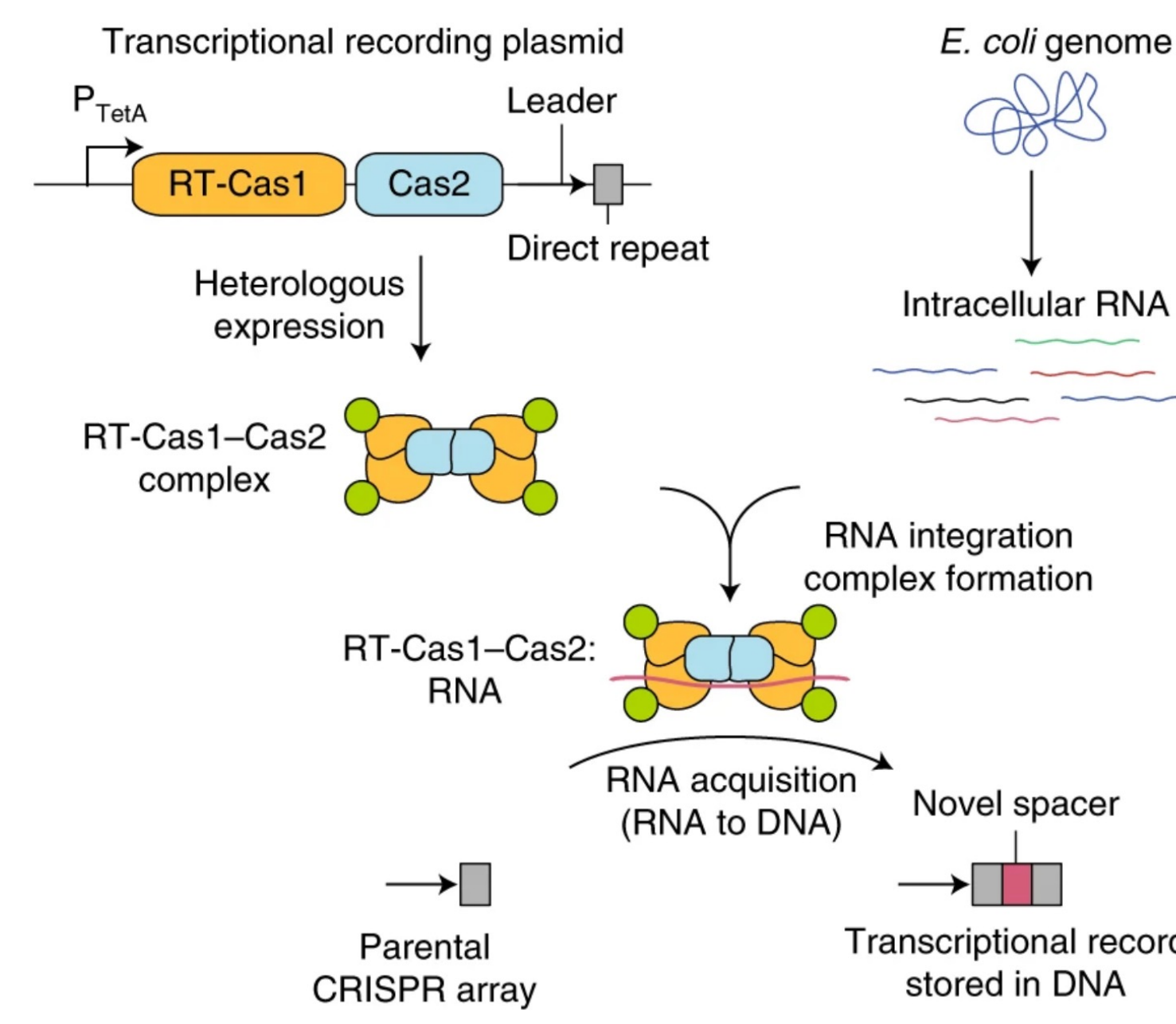
Bioengineering Building Blocks

CRISPR¹ (Clustered Regularly Interspaced Palindromic Repeats)



Can **collect** phage DNA for future immune responses

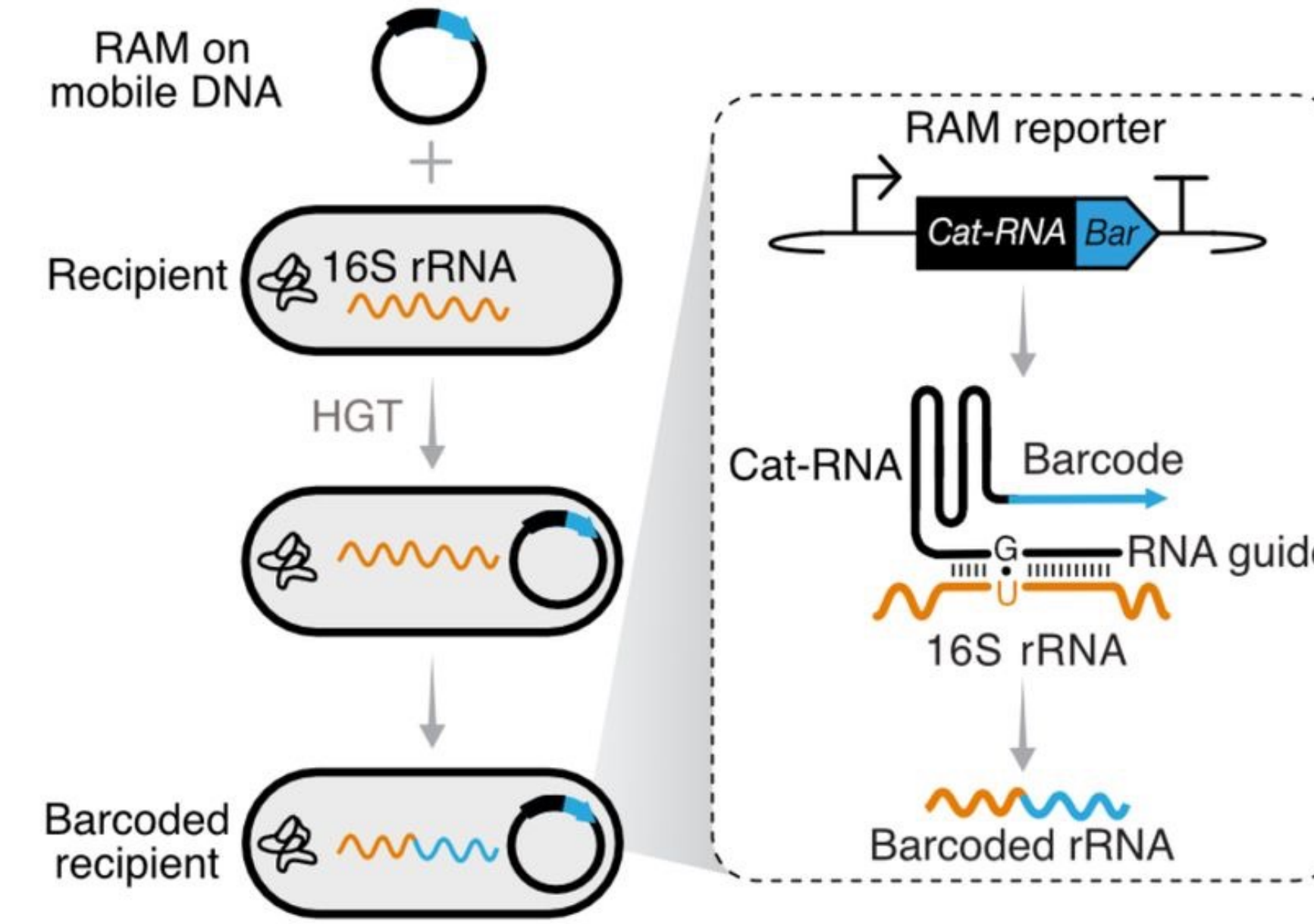
Record-Seq²



Can **record** RNAs of the transcriptome

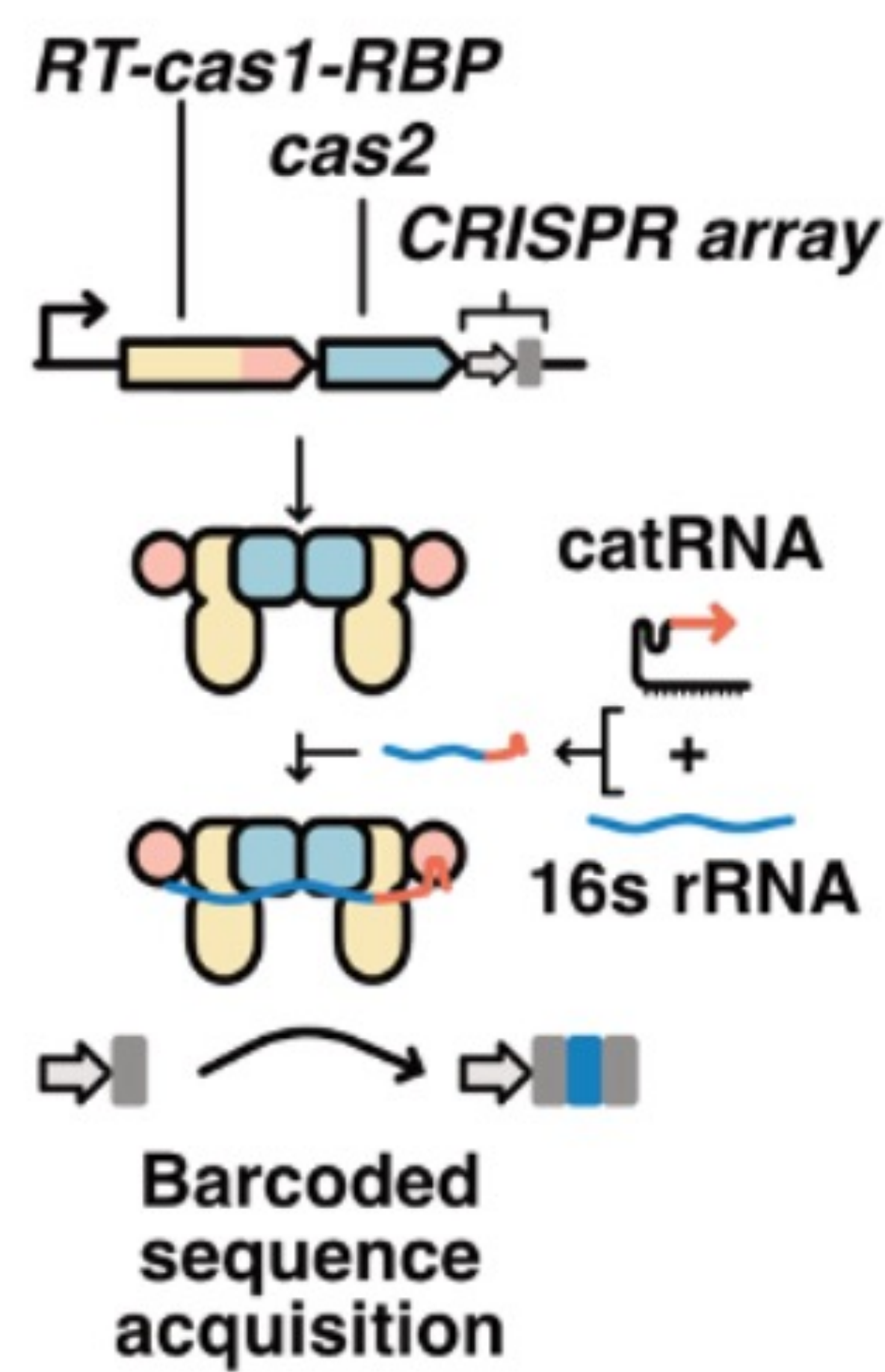
RAM³

(RNA-Addressable
Modification)



Can **identify** microbes that received a plasmid

Our Proposal



Cat-RNA adds a barcode denoting an HGT event to a non-conserved 16s region (unique identifier of host organism);

RT-Cas reverse transcribes HGT barcode + 16s region into CRISPR array;

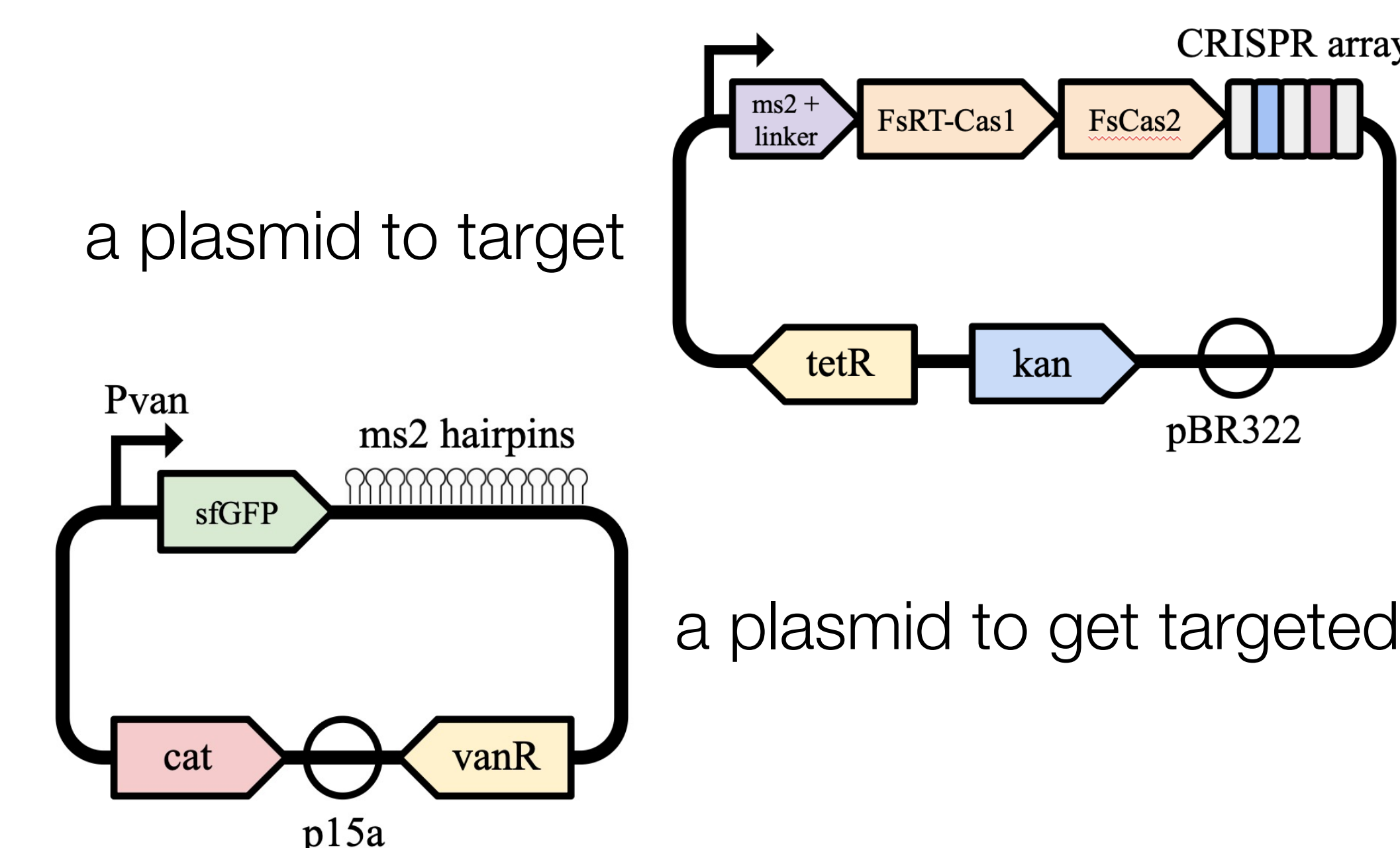
as time passes, the CRISPR array accumulates unique barcoded 16s signatures;

resulting in a recorded history of plasmid transfer across microbial hosts

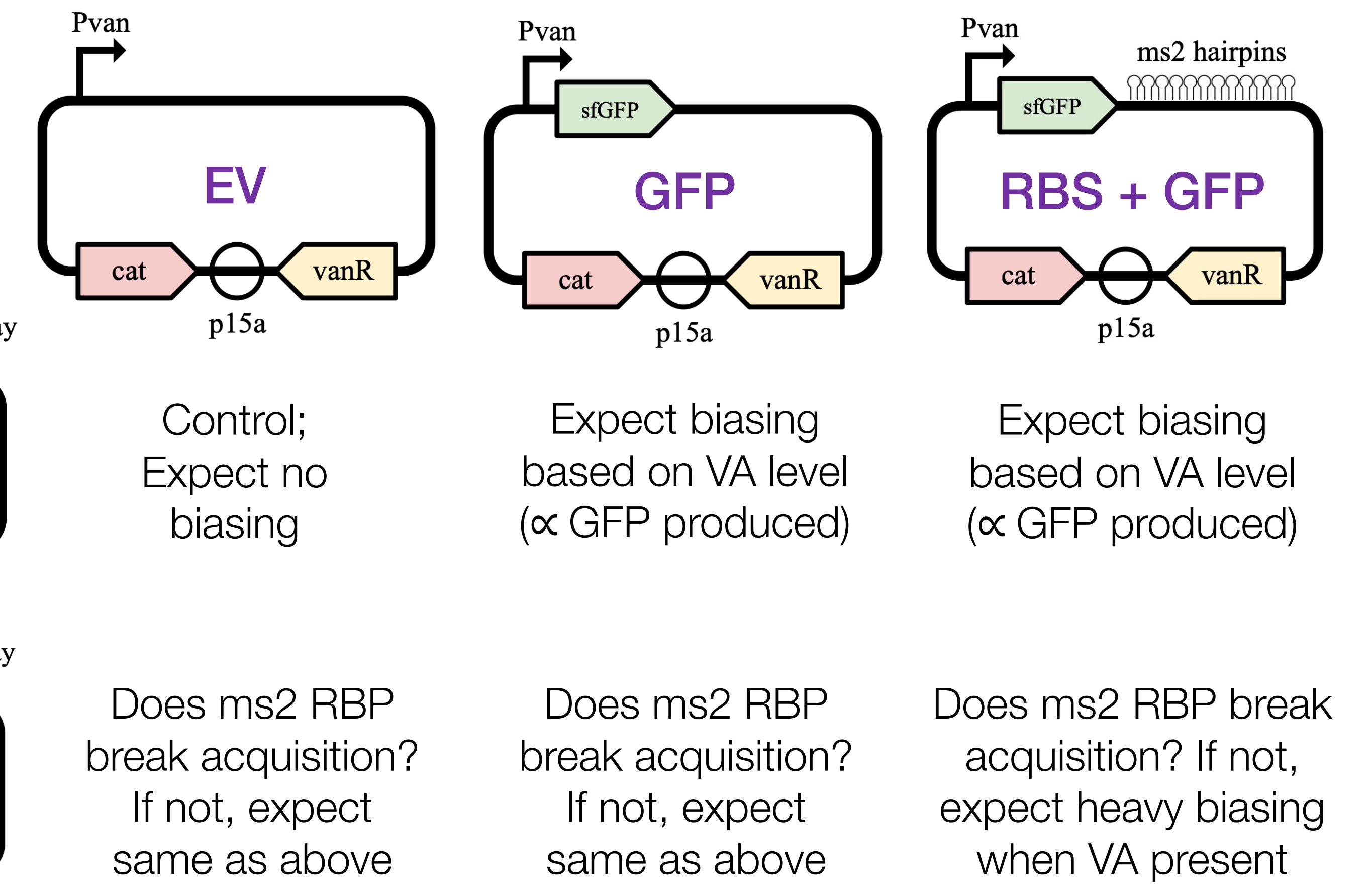
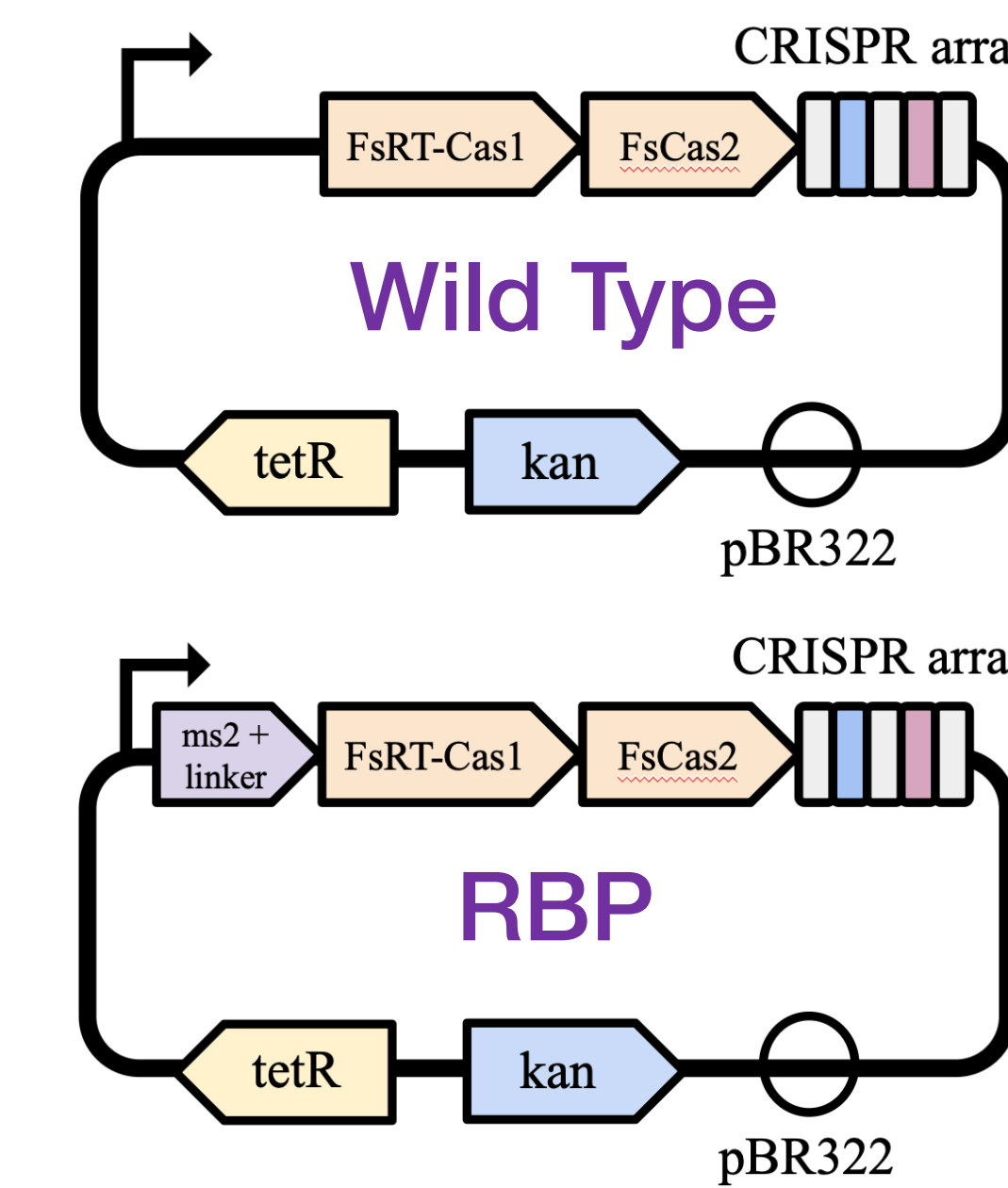
A Starting Point

Guiding Question: can we bias the CRISPR array to contain a target sequence by recreating Record-Seq?

Proof of Concept: bias the CRISPR array to GFP using RNA-binding protein interactions (specifically the ms2 RBP and RBS)



Experimental Design

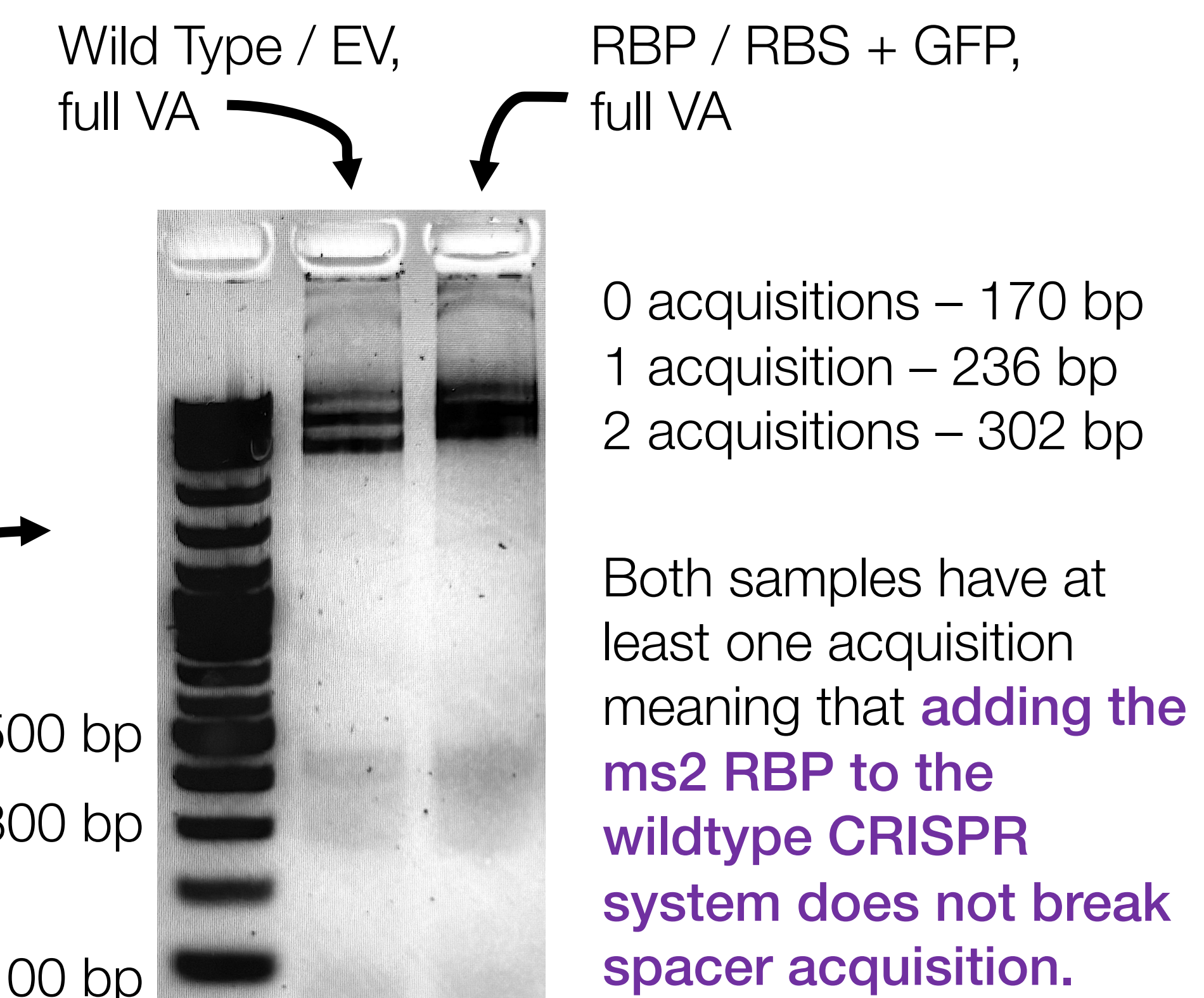


SENECA

Selective amplification of ExpaNdEd CRISPR Arrays

Quantifies RNA abundance based on spacer acquisition frequency

1. Spacer Acquisition
2. Golden Gate
3. First Round PCR
4. Second Round PCR
5. Sequencing
6. Transcriptional History Reconstruction



Next Steps

- Integrating RAM
- Toehold Switch Mechanism
 - Problem: ms2 RBP is faster at targeting than cat-RNA is at barcoding
 - Solution: ms2 RBS should be inactive while cat-RNA is barcoding

Acknowledgements

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References

- [1] Doudna JA. CRISPR in nature. In: Hochstrasser ML, et al., editors. CRISPRpedia [Internet]. Berkeley (CA): Innovative Genomics Institute, University of California, Berkeley; 2022 [updated 2022 Sep 12; cited 2026 Jul 26]. Available from: <https://innovativegenomics.org/crisprpedia/crispr-in-nature/>
- [2] Tanna T, Schmidt F, Cherepkova MY, Okoniewski M, Platt RJ. Recording transcriptional histories using Record-seq. Nat Protoc. 2020;15(2):513–539. doi:10.1038/s41596-019-0253-4.
- [3] Karinje L, Silberg JJ, Chappell J, Stadler LB. Rational engineering enhances the signal and modularity of an RNA barcoding technology to track gene transfer in microbiomes [preprint]. bioRxiv. 2026 Apr 21. doi:10.64898/2026.04.20.719664.